

Defecation physiology

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Learning Objectives

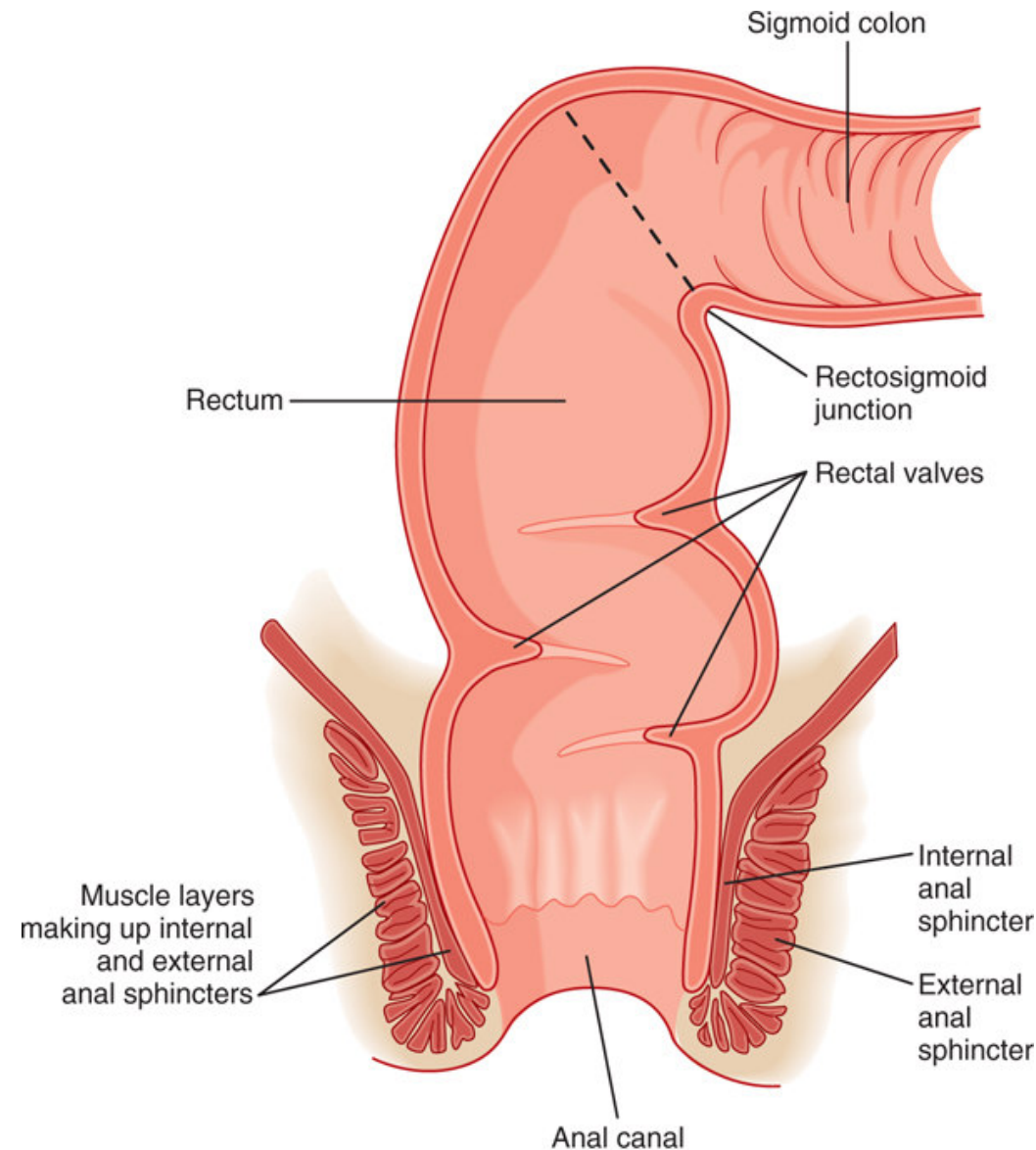
By the end of this lecture, students will be able to:

- Describe the anatomical and functional mechanisms that prevent continuous filling of the rectum.
- Differentiate between the internal and external anal sphincters according.
- Explain the intrinsic myenteric defecation reflex and its role in initiating defecation.
- Explain the parasympathetic defecation reflex, including the role of the sacral spinal cord and pelvic nerves.
- Compare the intrinsic and parasympathetic defecation reflexes in terms of their strength and effectiveness.
- Explain the abdominal and respiratory responses that assist defecation.

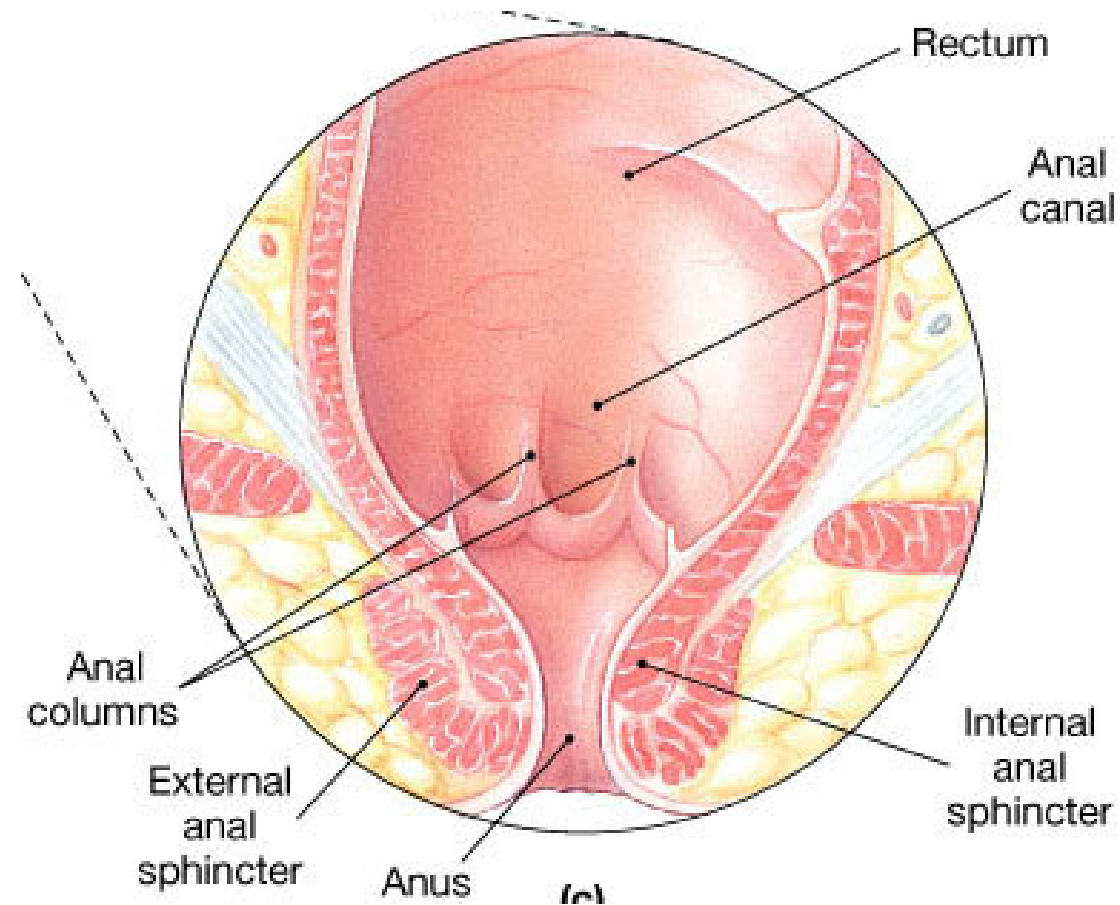
DEFECATION

The rectum is empty of feces most of the time because of :

- Weak functional sphincter exists about 20 cm from the anus at the juncture between the sigmoid colon and the rectum.
- Sharp angulation is also present here that contributes additional resistance to filling of the rectum.



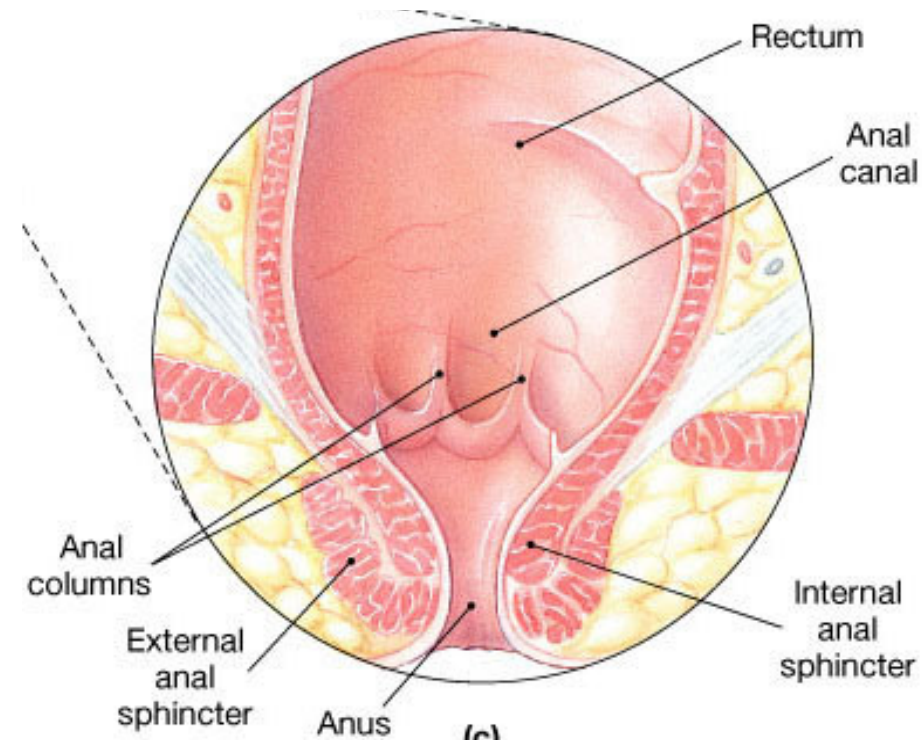
- When a mass movement forces feces into the rectum, the desire for defecation occurs immediately, including reflex contraction of the rectum and relaxation of the anal sphincters.



Continual dribble of fecal matter through the anus prevented by tonic constriction of the following:

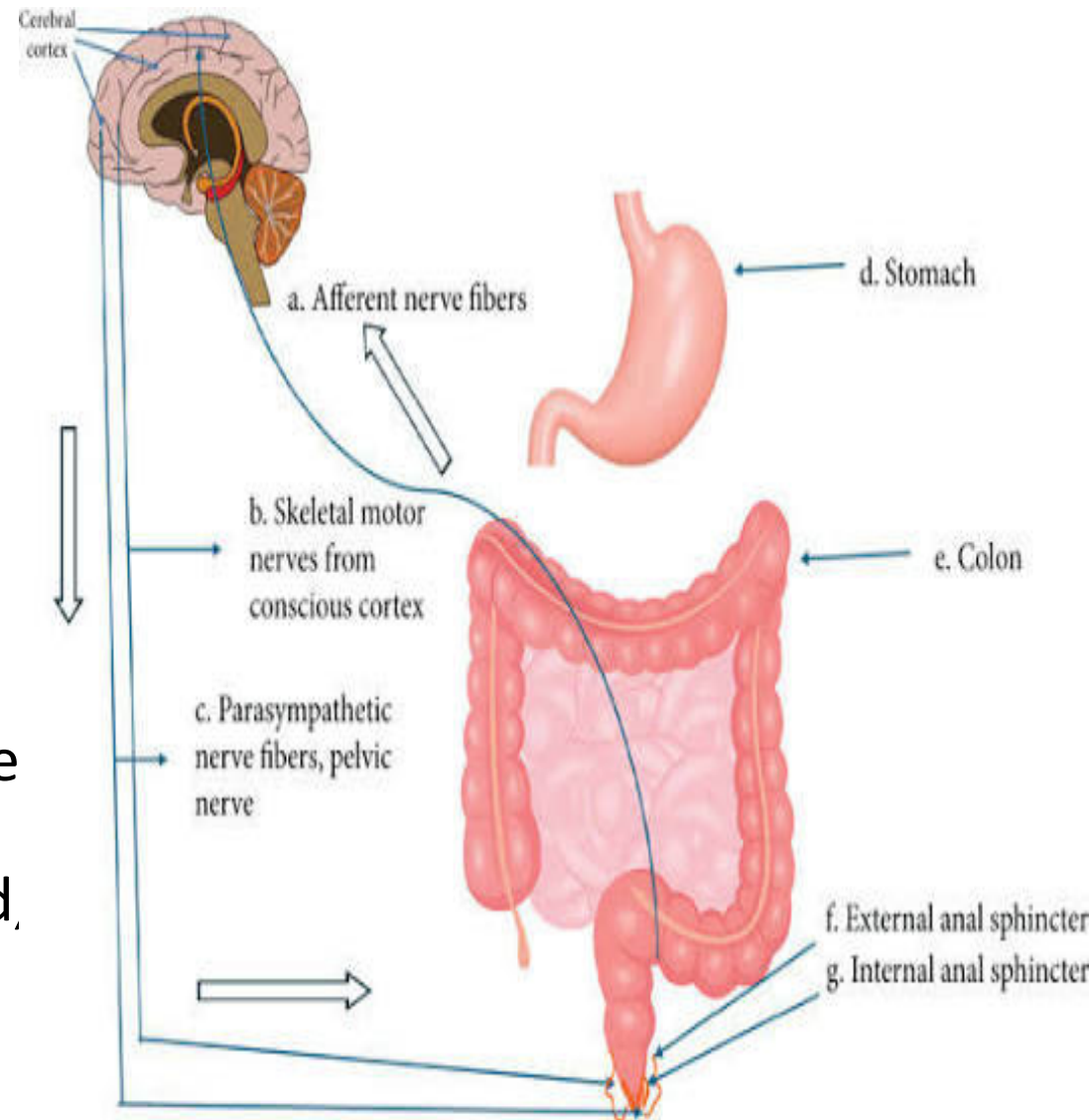
(1) Internal anal sphincter: is several cm long thickening of circular smooth muscle that lies inside the anus.

(2) External anal sphincter: striated voluntary muscle surrounds internal sphincter and extends distal to it. Controlled by nerve fibers in the **pudendal nerve**, part of the somatic nervous system, under voluntary, conscious, or subconscious control, continuously constricted unless conscious signals inhibit the constriction.



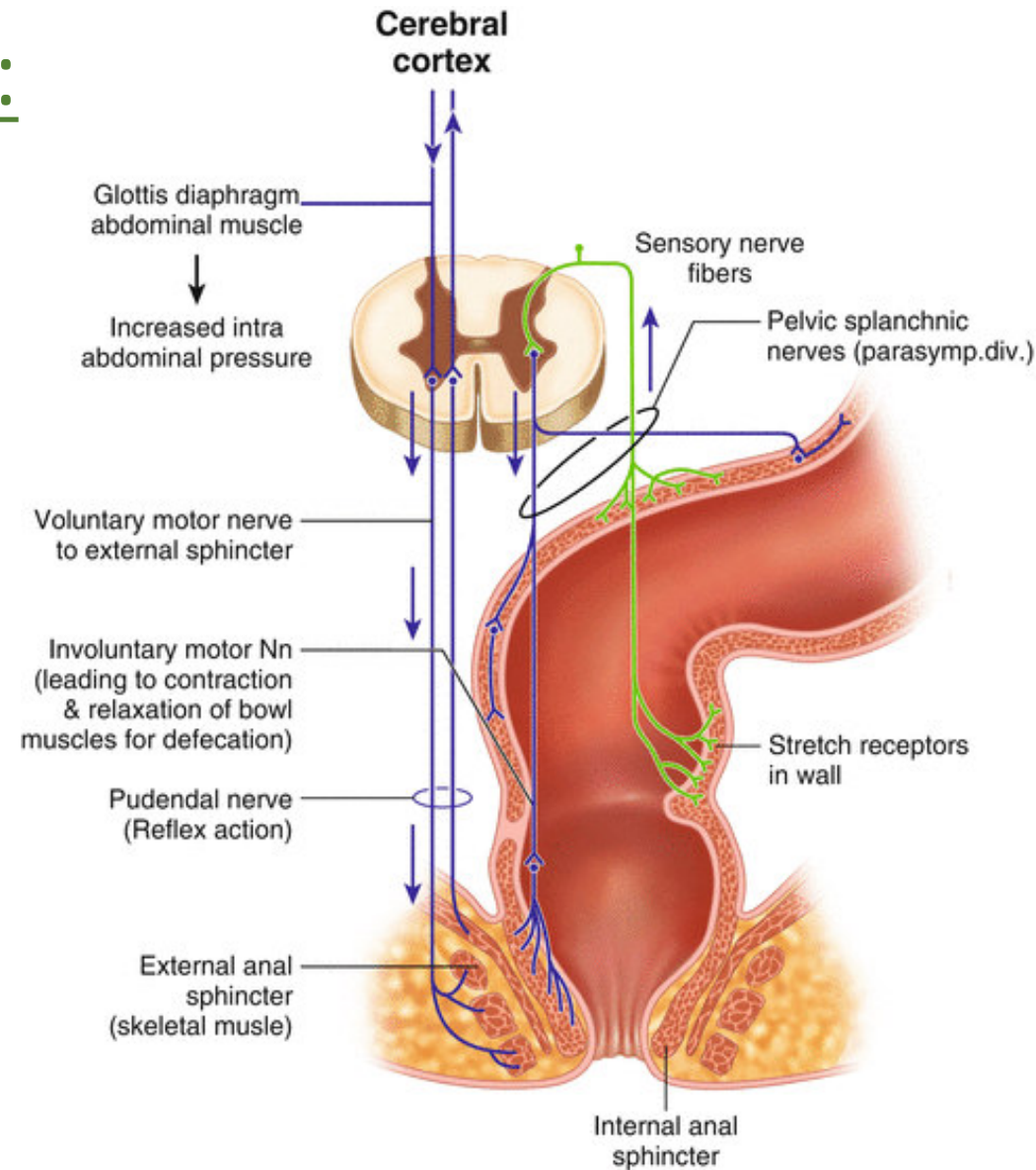
Defecation Reflexes:

- Defecation is initiated by defecation reflexes.
- One of these reflexes is an **intrinsic reflex** mediated by the local enteric nervous system in the rectal wall.
- When feces enter the rectum, distention of the rectal wall initiates afferent signals that spread through the myenteric plexus to initiate peristaltic waves in the descending colon, sigmoid, and rectum, forcing feces toward the anus.



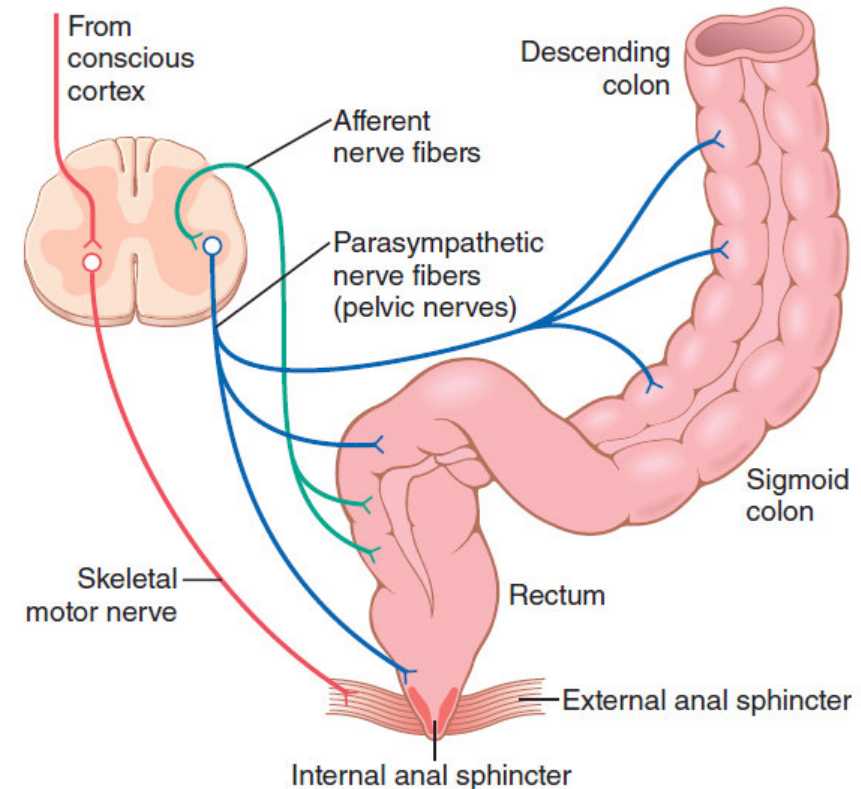
Defecation Reflexes cont.. :

- As the peristaltic wave approaches the anus, the internal anal sphincter is relaxed by inhibitory signals from the myenteric plexus; if the external anal sphincter is also consciously, voluntarily relaxed at the same time, defecation occurs.
- Normally when the intrinsic myenteric defecation reflex is functioning by itself, it is relatively weak.



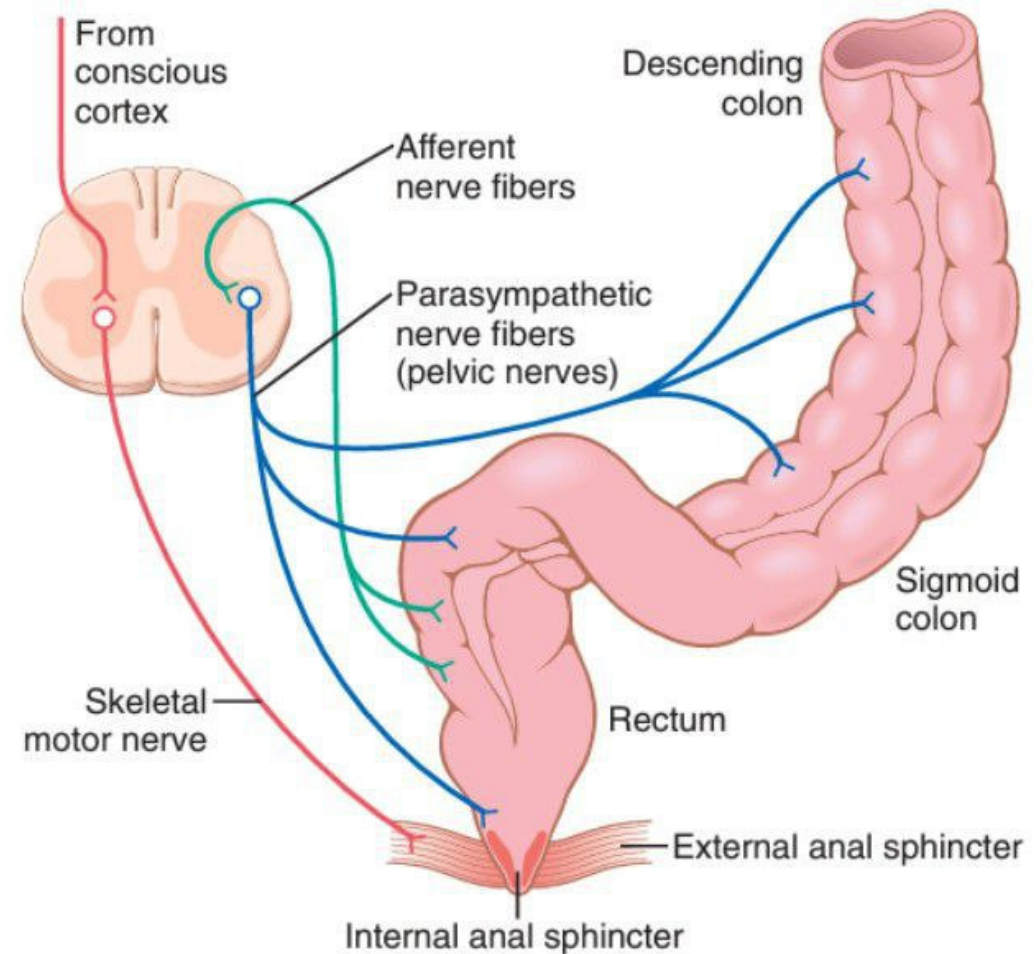
Defecation Reflexes cont.. :

- To be effective in causing defecation, fortified by another type of defecation reflex called a **parasympathetic defecation reflex** that involves the sacral segments of the spinal cord.
- When the nerve endings in the rectum stimulated, signals transmitted to the spinal cord and then back to the descending colon, sigmoid, rectum, and anus via parasympathetic nerve fibers in the pelvic nerves.



Defecation Reflexes cont.. :

- Parasympathetic signals intensify peristaltic waves and relax the internal anal sphincter, converting the intrinsic myenteric defecation reflex from a weak effort into a powerful process of defecation that is effective in emptying the large bowel from the splenic flexure of the colon to the anus.



Defecation Reflexes cont.. :

Defecation signals entering the spinal cord initiate other effects, such as

- taking a deep breath
- closure of the glottis
- contraction of the abdominal wall muscles to force the fecal contents of the colon downward
- Relax downward the pelvic floor and pull outward on the anal ring to evaginate the feces.

Defecation Reflexes cont.. :

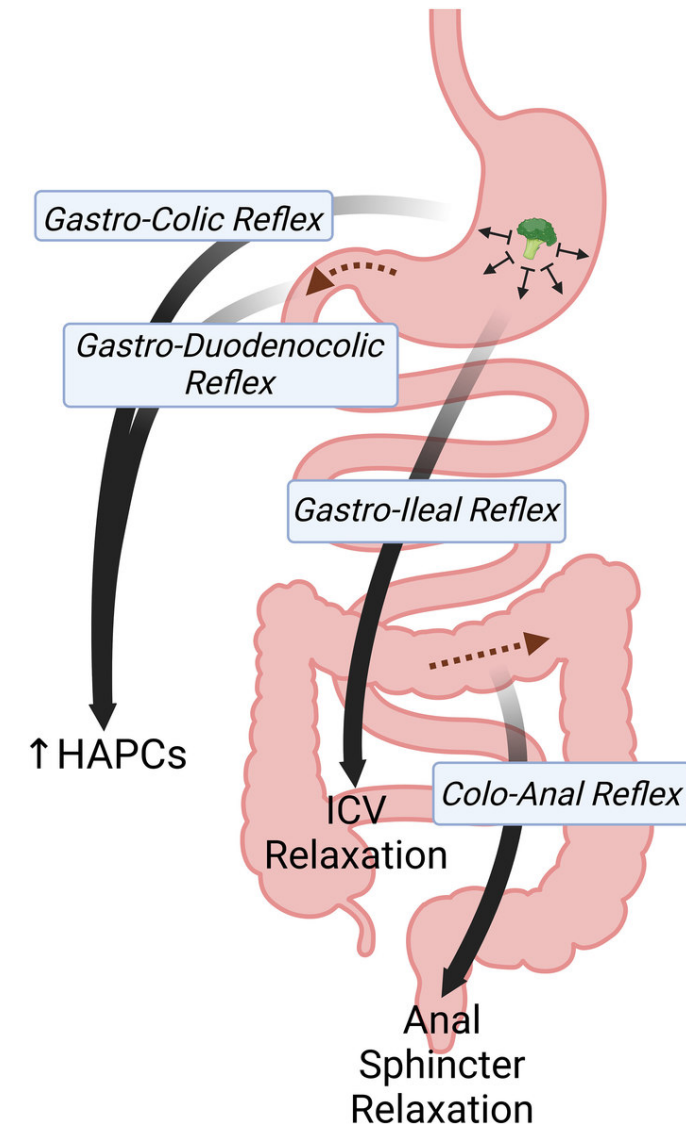
- When it becomes convenient for the person to defecate, the defecation reflexes can purposely be activated by taking a deep breath to move the diaphragm downward and contracting the abdominal muscles to increase the pressure in the abdomen, thus forcing fecal contents into the rectum to cause new reflexes.
- Reflexes initiated in this way are never as effective as those that arise naturally, and thus people who often inhibit their natural reflexes are likely to become constipated.

Defecation Reflexes cont.. :

- In newborns and in people with transected spinal cords, the defecation reflexes cause automatic emptying of the lower bowel because of lack of conscious control through voluntary contraction or relaxation of the external anal sphincter.
- The amount of bowel movements depends on various factors such as diet, health, and stress.
- The normal range of bowel activity varies from 2-3 bowel movements per day to 3-4 per week.

Other autonomic reflexes that affect bowel activity:

- Aside from the duodenocolic, gastrocolic, gastroileal, enterogastric, and defecation reflexes; other important nervous reflexes also can affect the overall degree of bowel activity.
- They are the **peritoneointestinal reflex**, **renointestinal reflex**, and **vesicointestinal reflex**.
- The peritoneointestinal reflex results from irritation of the peritoneum; inhibits the excitatory enteric nerves and thereby can cause intestinal paralysis, especially in patients with peritonitis.
- The renointestinal and vesicointestinal reflexes inhibit intestinal activity as a result of kidney or bladder irritation, respectively.



Diarrhoea:

- is an increase in the frequency, volume, and fluid content of the faeces caused by increased motility of and decreased absorption by the intestines.
- When chyme passes too quickly through the small intestine and faeces pass too quickly through the large intestine, there is not enough time for absorption.
- Frequent diarrhoea can result in dehydration and electrolyte imbalances.
- Excessive motility may be caused by lactose intolerance, stress, and microbes that irritate the gastrointestinal mucosa.

Constipation:

- Infrequent or difficult defecation caused by decreased motility of the intestines.
- Because the faeces remain in the colon for prolonged periods, excessive water absorption occurs, and the faeces become dry and hard.
- Constipation may be caused by poor habits (delaying defecation), spasms of the colon, insufficient fibre in the diet, inadequate fluid intake, lack of exercise, emotional stress, and certain drugs.
- A common treatment is a mild laxative, which induces defecation. However, that laxatives are habit-forming, and adding fibre to the diet, increasing amount of exercise, and increasing fluid intake are safer ways of controlling this problem.



THANK YOU
FOR YOUR
ATTENTION